Theorem 9.2

Let I be an open interval containing $x_0 \in \mathbb{R}$

Let $f, g : I \to \mathbb{R}$.

Suppose that c is a constant and the limits $\lim_{x \to x_0} f(x)$ and $\lim_{x \to x_0} g(x)$ exist. Then $\lim_{x \to x_0}(cf(x)), \lim_{x \to x_0}[f(x)+g(x)]$, $\lim_{x \to x_0}[f(x) - g(x)]$, $\lim_{x \to x_0}[f(x)g(x)]$ and $\lim_{x \to x_0} \frac{f(x)}{g(x)}, g(x) \neq 0$, all exist. Moreover,

(i) $\lim_{x \to x_0} cf(x) = c \lim_{x \to x_0} f(x)$,

(ii) $\lim_{x \to x_0}[f(x) \pm g(x)] = \lim_{x \to x_0} f(x) \pm \lim_{x \to x_0} g(x)$,

(iii) $\lim_{x \to x_0}[f(x).g(x)] = \lim_{x \to x_0} f(x) . \lim_{x \to x_0} g(x)$ and

(iv) $\lim_{x \to x_0} \frac{f(x)}{g(x)} = \frac{\lim_{x \to x_0} f(x)}{\lim_{x \to x_0} g(x)}$, provided $\lim_{x \to x_0} g(x) \neq 0$.

These results can be extended to any finite number of functions. ■

Example 9.9

Compute (i) : $\lim_{x \to 8}(5x)$ (ii) $\lim_{x \to -2}\left(-\frac{3}{2}x\right)$.

Solution

(i) $\lim_{x \to 8}(5x) = 5\lim_{x \to 8}(x) = 5 \times 8 = 40$.

(ii) $\lim_{x \to -2}\left(-\frac{3}{2}x\right) = -\frac{3}{2}\lim_{x \to -2}(x) = \left(-\frac{3}{2}\right)(-2) = 3$.

Example 9.10

Compute $\lim_{x \to 0}\left[\frac{x^2+x}{x} + 4x^3 + 3\right]$.

Solution

$$\lim_{x \to 0}\left[\frac{x^2+x}{x} + 4x^3 + 3\right] = \lim_{x \to 0}\left(\frac{x^2+x}{x}\right) + \lim_{x \to 0}(4x^3+3)$$

$$= \lim_{x \to 0}(x+1) + \lim_{x \to 0}(4x^3+3)$$

$$= (0+1) + (0+3)$$

$$= 4.$$

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